

Home Lab Project Documentation week 4 day 2

Review basic Linux command-line operations and navigation to prepare for cybersecurity analysis activities within the Kali Linux environment.

For **Week 4, Day 2**, this task is focused on becoming comfortable with the **Kali Linux command line** before you start more hands-on cybersecurity analysis.

Week 4 — Day 2

Task: Review basic Linux command-line operations and navigation to prepare for cybersecurity analysis activities within the Kali Linux environment.

You don't need to do any penetration testing today. The goal is to practice the commands you'll use later for **network analysis, reconnaissance, file management, and security tools**.

What to practice in Kali

Area	Commands to Practice	Purpose
Current location	<code>pwd</code>	Show your current directory
List files	<code>ls</code> , <code>ls -la</code>	View files and directories
Navigate	<code>cd</code> , <code>cd ..</code> , <code>cd ~</code>	Move between directories
Create directory	<code>mkdir</code>	Create a working directory
Create file	<code>touch</code>	Create an empty file
Copy	<code>cp</code>	Copy files/directories
Move/Rename	<code>mv</code>	Move or rename files
Delete	<code>rm</code>	Remove files
View files	<code>cat</code> , <code>less</code>	Read file contents
Search	<code>find</code> , <code>grep</code>	Locate files or text

System information	<code>whoami</code> , <code>hostname</code> , <code>uname -a</code>	Identify the current system/user
Network information	<code>ip addr</code> , <code>ip route</code>	Review network configuration
Connectivity	<code>ping</code>	Test network connectivity
Command help	<code>man</code> , <code>--help</code>	Learn command usage

Recommended hands-on exercise

In Kali, create a small practice directory:

```
mkdir ~/cybersecurity-practice
cd ~/cybersecurity-practice
```

Check where you are:

```
pwd
```

Create a practice file:

```
touch notes.txt
```

List it:

```
ls -la
```

Add some information:

```
echo "Kali Linux cybersecurity lab" > notes.txt
```

View it:

```
cat notes.txt
```

Then review your network configuration:

```
ip addr
ip route
```

Since your lab uses the **192.168.64.0/24** isolated network, you should be able to identify Kali's `192.168.64.2` address and the route associated with your virtual network.

What to document for Day 2

You can document:

- Basic Linux directory navigation
- File and directory management
- Viewing and searching files
- Basic system-identification commands
- Basic network commands
- Help/manual commands
- Confirmation that the commands were successfully practiced in Kali

How Day 2 fits with Day 1

There is **no significant repetition**:

- **Day 1:** Verify the virtual network and establish VM recovery snapshots.
- **Day 2:** Build Linux command-line skills inside Kali.
- **Day 3:** Install/configure Wireshark and prepare for packet capture.
- **Day 4:** Practice Wireshark interfaces, filters, coloring rules, and packet capture.

So Day 2 is essentially the **Linux command-line preparation day** before you begin the actual Wireshark/network-analysis work.